

****These questions are basic questions to get you started with your studying.****

Chapter 2 Practice problems

1. What are the four essential elements of Life?
 - a. Nitrogen, Hydrogen, Oxygen & Helium
 - b. Hydrogen, Nitrogen, Carbon, & Fluorine
 - c. Hydrogen, Oxygen, Carbon, & Nitrogen
 - d. Sodium, Nitrogen, Phosphorous & Sulfur
2. What is the mass number of an atom?
 - a. Protons + electrons
 - b. Electrons + neutrons
 - c. Only Protons
 - d. Protons + neutrons
3. The number of _____ differs in isotopes
 - a. Neutrons
 - b. Protons
 - c. Electrons
4. An electron in which shell will have the highest potential energy?
 - a. 1st shell
 - b. 2nd shell
 - c. 3rd shell
5. How many electrons will you find in the 3rd shell of an Aluminum (Al) atom (atomic # = 13)
 - a. 8
 - b. 3
 - c. 6
 - d. 2
6. Draw the electron shell diagram for Phosphorous (atomic # = 15), Identify the number of valence electrons, the valence number, and the number of bonds it will form.
7. What does it mean for an atom to have a full valence shell? How will it interact with other atoms?
8. What is the result of an ionic bond between Na & Cl?
 - a. A double bond
 - b. Two molecules that are attracted to each other due to resulting charges
 - c. A sodium anion & a chlorine cation
 - d. Transferring of electrons
 - e. Both b & d are correct
9. What characteristics below are those of a hydrogen bond?
 - a. Occurs between water molecules and other polar molecules

- b. Extremely important to the specific properties of water
 - c. A bond of molecular force
 - d. All of the above
10. What does it mean for a molecule to be polar?
11. Which of the following is the most polar molecule?
- a. HF
 - b. CO₂
 - c. F₂
 - d. NO₂
12. Magnesium has an atomic number of 25 and a mass number of 55. Which of the following is true?
- a. Magnesium has 25 electrons
 - b. Magnesium has 25 protons
 - c. Magnesium has 30 neutrons
 - d. None of the above are true
 - e. All of the above are true
13. Explain the difference between ionic bonds and covalent bonds
14. Lithium has a valence number of 1 and often donates that electron to another atom. The atom that received the electron would now be a(n):
- a. Cation
 - b. Anion
 - c. Isotope
 - d. Neutral
15. In the example above, after it donates an electron, Lithium is a(n):
- a. Cation
 - b. Anion
 - c. Isotope
 - d. Neutral
16. What happens if an atom gains or loses a proton?
- a. It becomes an isotope
 - b. It is no longer neutral
 - c. It will also gain or lose an electron
 - d. It is a different atom.

Answers:

- 1. C
- 2. D
- 3. A

4. C
5. B
6. 5 valence electrons, valence number = 3, will form 3 bonds. Shell 1 will be full with 2 electrons, shell 2 will be full with 8 electrons, and the third shell will have 5 electrons, three of those are unpaired.
7. No bonds will form
8. E
9. D
10. One atom has more electronegativity than other so the electrons spend more time around it than they do around the less electronegative atom. This results in one side of the molecule having a partial positive charge and the other side having a partial negative charge.
11. A
12. E
13. In ionic bonds atoms gain or lose electrons whereas in covalent bonds, atoms share electrons
14. B
15. A
16. D